

Reflection Essay on the Single View Geometry and Distance Estimation Assignment

This assignment on Single View Geometry and Distance Estimation gave me a deeper understanding of how computer vision and projective geometry can be applied to solve real-world problems. Before working on this task, I mainly associated object detection systems such as YOLO with identifying objects inside images. However, through this assignment I learned that computer vision is not only about detecting objects, but also about understanding spatial relationships and estimating measurements from a single image using geometric principles.

One of the most interesting parts of the assignment was estimating the distance between the bicycle and the IIT Bombay arch using cross-ratio geometry. I found it fascinating that it is possible to estimate real-world distances from only one perspective image when appropriate reference points and vanishing points are chosen correctly. The assignment demonstrated how concepts from mathematics and geometry directly connect with modern AI-based computer vision systems.

The assignment also improved my understanding of perspective projection and vanishing points. I realized that even small changes in manually selected points can significantly affect the final distance estimate. Choosing the correct image points such as A' , B' , C' , and V' was therefore extremely important. In particular, selecting the bottom-center of the YOLO bounding box as the bicycle reference point highlighted how geometry and practical assumptions work together. I learned that the bottom-center point is more meaningful because it approximates the object's contact with the ground plane.

Another valuable learning experience came from using YOLO object detection. Running pretrained YOLO models and understanding COCO class IDs helped me understand how deep learning models can automatically identify objects like bicycles and people in images. At the same time, I observed practical limitations of object detection. For example, when the assignment image already contained annotations, the model sometimes struggled to rediscover the bicycle because parts of the image were visually blocked. This taught me that machine learning systems are highly dependent on image quality and visibility.

The assignment also strengthened my technical skills in Python, OpenCV, and PyTorch. I gained practical exposure to working with scripts, handling JSON data, and organizing

reproducible workflows. Setting up the environment on native Apple Silicon hardware was another important experience because it showed me how hardware acceleration such as Apple's MPS backend can improve AI model execution. Understanding environment configuration, dependencies, and GPU support is an important skill for real-world AI development.

The personal-photo experiment was another meaningful part of the assignment because it connected the theory with a real-life scenario. Measuring the camera-to-person distance physically and comparing it with what could theoretically be estimated through projective geometry made me think critically about when mathematical estimation methods are reliable and when direct measurement is more practical. This helped me understand that engineering decisions often involve balancing theory with practicality.

One challenge I faced during the assignment was understanding the sensitivity of the cross-ratio method. Small variations in the vanishing point or bounding-box reference point changed the estimated distance noticeably. This made me appreciate the importance of careful data selection and precision in computer vision tasks. I also understood that assumptions in geometry-based estimation methods must always be clearly stated because the final result depends heavily on them.

Overall, this assignment was an excellent combination of mathematics, programming, and artificial intelligence. It allowed me to apply theoretical geometric concepts in a practical computer vision problem while also learning about object detection systems and machine learning workflows. More importantly, it improved my problem-solving approach by teaching me to verify assumptions, analyze sources of error, and think critically about the reliability of computational results. I believe the knowledge and experience gained from this assignment will be valuable for future work in computer vision, AI, and software development.